

Lab 11: Investigating browser history

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Image “cfreds_2015_data_leakage_pc.dd” from Lab 5

Goals

- Understand the application layer of a computer system that generates evidence
- Understand the behavioral model of an application
- The approach to investigating browser history (evidence generated by an application)
- Familiar with important tools for browser investigations

Step 1

- Set up loop device (a pseudo-**device** that makes a file accessible as a block **device**)

```
losetup --partscan --find --show --read-only cfreds_2015_data_leakage_pc.dd
```

```
mkdir /mnt/loopdev
```

```
mount -o ro,loop,offset=206848 cfreds_2015_data_leakage_pc.dd /mnt/loopdev
```

```
(root@hainguyen)-[/home/hainguyen]
# cd lab

(root@hainguyen)-[/home/hainguyen/lab]
# losetup --partscan --find --show --read-only cfreds_2015_data_leakage_
pc.dd
/dev/loop0
```

- Check all versions of IE via .log

```
(root@hainguyen)-[/home/hainguyen/lab]
# ls /media/hainguyen/C8CA0C8DCA0C7A48/Windows/*.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/DtcInstall.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/IE11_main.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/PFR0.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/setupact.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/setuperr.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/TSSysprep.log
/media/hainguyen/C8CA0C8DCA0C7A48/Windows/WindowsUpdate.log

(root@hainguyen)-[/home/hainguyen/lab]
#
```

- Exam IE logs

```
(root@hainguyen)-[/home/hainguyen/Lab]
# cat /media/hainguyen/C8CA8C8DCA8C7A48/Windows/IE11_main.log | head
00:00.000: =====
00:00.000: Started: 2015/03/22 (Y/M/D) 11:12:32.577 (local)
00:00.000: Time Format in this log: MM:ss.mmm (minutes:seconds.milliseconds)
00:00.000: Command line: "C:\Users\informant\Desktop\Download\IE11-Windows6.1-x64-en-us.exe"
00:00.000: INFO: Setup installer for Internet Explorer: 11.0.9600.16428
00:00.000: INFO: Previous version of Internet Explorer: 8.0.7601.17514
00:00.015: INFO: Checking if iexplore.exe's current version is between 11.0.9600.0...
00:00.015: INFO: ...and 11.1.0.0...
00:00.015: INFO: Maximum version on which to run IEAK branding is: 11.1.0.0...
00:00.015: INFO: iexplore.exe version check success. Install can proceed.

(root@hainguyen)-[/home/hainguyen/Lab]
#
```

- Can you find the version number?

Use Windows Registry hive shell (*hivexsh*)

- Install *hivexsh*

```
(root@hainguyen)-[/home/hainguyen/Lab]
# sudo apt-get install libhivex-bin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libhivex-bin is already the newest version (1.3.20-1+b10).
The following packages were automatically installed and are no longer required:
  firebird3.0-common firebird3.0-common-doc icu-devtools libabsl20230802
  libbfiol1 libc++1-19 libc++abi1-19 libcapstone4 libconfig++9v5 libconfig9
  libdirectfb-1.7-7t64 libegl-dev libflac12t64 libfmt9 libfuse3-3
  libgeos3.13.0 libgl1-mesa-dev libglapi-mesa libgles-dev libgles1
  libglvnd-core-dev libglvnd-dev libgtksourceview-3.0-1
  libgtksourceview-3.0-common libgtksourceviewmm-3.0-0v5 libgumbo2
  libicu-dev libjxl0.9 libmbcrypto7t64 libmsgpack-0-1 libopenh264-7
  libpaper1 libpython3.12-dev libpython3.12-minimal libpython3.12-stdlib
  libpython3.12t64 libqt5sensors5 libqt5webkit5 libsuperlu6 libtag1v5
  libtag1v5-vanilla libtagc0 libunwind-19 libutempter0
  libwebpct-audio-processing1 libx265-200 openjdk-23-jre
  openjdk-23-jre-headless python3-appdirs python3-dunamai
  python3-nfsclient python3-ntlm-auth python3-packaging-whl
  python3-poetry-dynamic-versioning python3-requests-ntlm
  python3-setproctitle python3-tomlkit python3-wheel-whl python3.12
  python3.12-dev python3.12-minimal python3.12-venv ruby-zeitwerk ruby3.1
  ruby3.1-dev ruby3.1-doc strongswan
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 244 not upgraded.

(root@hainguyen)-[/home/hainguyen/Lab]
#
```

- Exam the version of IE using *hivexsh*

```
(root@hainguyen)-[/home/hainguyen/Lab]
# hivexsh /media/hainguyen/C8CA8C8DCA8C7A48/Windows/System32/config/SOFTWARE

Welcome to hivexsh, the hivex interactive shell for examining
Windows Registry binary hive files.

Type: 'help' for help summary
      'quit' to quit the shell

SOFTWARE\> cd Microsoft\Internet Explorer
SOFTWARE\Microsoft\Internet Explorer> lsval svcVersion
11.0.9600.17691
SOFTWARE\Microsoft\Internet Explorer> |
```

Step 2.

Identify directory/file paths related to the web browser history

MS IE (9 or lower)	C:\Users\informant\AppData\Local\Microsoft\Windows\History\ C:\Users\informant\AppData\Local\Microsoft\Windows\Temporary Internet Files\Low\Content.IE5\index.dat C:\Users\informant\AppData\Roaming\Microsoft\Windows\Cookies\
MS IE 11	C:\Users\informant\AppData\Local\Microsoft\Windows\WebCache\WebCacheV01.dat

– Internet Explorer 11 History

```
(root@hainguyen)-[/home/hainguyen/Lab]
# ls -l /media/hainguyen/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/WebCache
total 35400
-rwxr-xr-x 1 hainguyen hainguyen 524288 Mar 25 2015 V0100024.log
-rwxr-xr-x 1 hainguyen hainguyen 524288 Mar 25 2015 V0100025.log
-rwxr-xr-x 1 hainguyen hainguyen 8192 Mar 25 2015 V01.chk
-rwxr-xr-x 1 hainguyen hainguyen 524288 Mar 25 2015 V01.log
-rwxr-xr-x 1 hainguyen hainguyen 524288 Mar 22 2015 V01res00001.jrs
-rwxr-xr-x 1 hainguyen hainguyen 524288 Mar 22 2015 V01res00002.jrs
-rwxr-xr-x 1 hainguyen hainguyen 33619968 Mar 25 2015 WebCacheV01.dat
(root@hainguyen)-[/home/hainguyen/Lab]
#
```

– Find Google Chrome Cache, Cookies, and History

```
root@hainguyen: /home/hainguyen x + v
(root@hainguyen)-[/home/hainguyen/Lab]
# ls -l /media/hainguyen/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Google/Chrome/User\ Data/Default/ | grep -Ei "Cache|Cookies|History"
drwxr-xr-x 1 hainguyen hainguyen 0 Mar 22 2015 Application Cache
drwxr-xr-x 1 hainguyen hainguyen 98304 Mar 24 2015 Cache
-rwxr-xr-x 1 hainguyen hainguyen 14212730 Mar 22 2015 ChromeDWriteFontCach
-rwxr-xr-x 1 hainguyen hainguyen 137216 Mar 24 2015 Cookies
-rwxr-xr-x 1 hainguyen hainguyen 16384 Mar 24 2015 Cookies-journal
-rwxr-xr-x 1 hainguyen hainguyen 6144 Mar 24 2015 Extension Cookies
-rwxr-xr-x 1 hainguyen hainguyen 1544 Mar 24 2015 Extension Cookies-journal
drwxr-xr-x 1 hainguyen hainguyen 0 Mar 23 2015 GPUCache
-rwxr-xr-x 1 hainguyen hainguyen 135168 Mar 24 2015 History
-rwxr-xr-x 1 hainguyen hainguyen 16384 Mar 24 2015 History-journal
-rwxr-xr-x 1 hainguyen hainguyen 47175 Mar 24 2015 History Provider Cache
drwxr-xr-x 1 hainguyen hainguyen 4096 Mar 24 2015 Media Cache
(root@hainguyen)-[/home/hainguyen/Lab]
#
```

What websites were the suspect accessing? (Timestamp, URL...)

- Get browser history files
- View these files

Step 1.

Copy Three Browser History Files

Copy IE 11 History

```
(root@hainguyen)-[/home/hainguyen/Lab]
# mkdir webhistory
(root@hainguyen)-[/home/hainguyen/Lab]
#
(root@hainguyen)-[/home/hainguyen/Lab]
# cp /media/hainguyen/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/WebCache/WebCacheV01.dat webhistory
(root@hainguyen)-[/home/hainguyen/Lab]
#
(root@hainguyen)-[/home/hainguyen/Lab]
# ls -l webhistory
total 32832
-rwxr-xr-x 1 root root 33619968 Jun 27 04:02 WebCacheV01.dat
(root@hainguyen)-[/home/hainguyen/Lab]
#
```

Copy IE 8 History

```
(root@hainguyen)-[/home/hainguyen/Lab]
# cp /media/hainguyen/C8CA0C8DCA0C7A48/Users/informant/AppData/Local/Microsoft/Windows/Temporary\ Internet\ Files/Low/Content.IE5/index.dat webhistory
(root@hainguyen)-[/home/hainguyen/Lab]
#
(root@hainguyen)-[/home/hainguyen/Lab]
# ls -l webhistory/index.dat
-rwxr-xr-x 1 root root 245760 Jun 27 04:05 webhistory/index.dat
(root@hainguyen)-[/home/hainguyen/Lab]
#
```

Copy Chrome History

```
root@hainguyen: /home/hain
#
(root@hainguyen)-[/home/hainguyen/lab]
# cp /media/hainguyen/CBCA0C8DCA0C7A48/Users/informant/AppData/Local/Google/Chrome/User\ Data/Default/History webhistory
#
(root@hainguyen)-[/home/hainguyen/lab]
# ls -l webhistory/History
-rwxr-xr-x 1 root root 135168 Jun 27 04:06 webhistory/History
#
(root@hainguyen)-[/home/hainguyen/lab]
#
```

Step 2.

View IE 11 History Using *libesedb*

Find the file type of *WebCacheV01.dat*

```
(root@hainguyen)-[/home/hainguyen/lab]
# file webhistory/WebCacheV01.dat
webhistory/WebCacheV01.dat: Extensible storage engine DataBase, version 0x620, checksum 0xe172001b, page size 32768, Windows version 6.1
(root@hainguyen)-[/home/hainguyen/lab]
# |
```

- Install *libesedb*

```
(root@hainguyen)-[/home/hainguyen/lab]
# sudo apt-get install libsecdb-utils -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  firebird3.0-common firebird3.0-common-doc icu-devtools
  libabsl20230802 libbfiol libc++1-19 libc++abi1-19 libcapstone4
  libconfig++v5 libconfig9 libdirectfb-1.7-7t64 libegl-dev
  libflac12t64 libfmt9 libfuse3-3 libgeo3.13.0 libglib-mesa-dev
```

- Export *WebCacheV01.dat* to the folder `webhistory/IE11`

```
root@hainguyen:/home/hainguyen/hain #
root@hainguyen:/home/hainguyen/lab #
root@hainguyen:/home/hainguyen/lab # esedbexport webhistory/WebCacheV01.dat -t webhistory/IE11
esedbexport 20240428

Opening file.
Database type: MSIE WebCache.
Exporting table 1 (MSysObjects) out of 37.
Exporting table 2 (MSysObjectsShadow) out of 37.
Exporting table 3 (MSysIncodeFixupV2) out of 37.
Exporting table 4 (Containers) out of 37.
Exporting table 5 (Container_1) out of 37.
Exporting table 6 (Container_2) out of 37.
Exporting table 7 (Container_3) out of 37.
Exporting table 8 (Container_4) out of 37.
Exporting table 9 (Container_5) out of 37.
Exporting table 10 (Container_6) out of 37.
Exporting table 11 (Container_7) out of 37.
Exporting table 12 (Container_9) out of 37.
Exporting table 13 (Container_10) out of 37.
Exporting table 14 (Container_11) out of 37.
Exporting table 15 (Container_12) out of 37.
Exporting table 16 (Partitions) out of 37.
Exporting table 17 (AppCacheEntry_1) out of 37.
Exporting table 18 (DependencyEntry_2) out of 37.
Exporting table 19 (Container_14) out of 37.
Exporting table 20 (Container_15) out of 37.
Exporting table 21 (LeakFiles) out of 37.
Exporting table 22 (AppCache_1) out of 37.
Exporting table 23 (AppCache_2) out of 37.
Exporting table 24 (Container_15) out of 37.
Exporting table 25 (Container_16) out of 37.
Exporting table 26 (AppCacheEntry_3) out of 37.
Exporting table 27 (Container_18) out of 37.
Exporting table 28 (DependencyEntry_4) out of 37.
Exporting table 29 (Container_19) out of 37.
Exporting table 30 (Container_21) out of 37.
Exporting table 31 (Container_17) out of 37.
Exporting table 32 (AppCache_3) out of 37.
Exporting table 33 (AppCache_4) out of 37.
Exporting table 34 (Container_22) out of 37.
Exporting table 35 (Container_23) out of 37.
Exporting table 36 (Container_24) out of 37.
Exporting table 37 (Container_25) out of 37.
Export completed.
root@hainguyen:/home/hainguyen/lab #
```

- Find the type of the file

```
(root@hainguyen)-[/home/hainguyen/lab]
# file webhistory/IE11.export/AppCache_1.21
webhistory/IE11.export/AppCache_1.21: ASCII text

(root@hainguyen)-[/home/hainguyen/lab]
# |
```

- Create three lines with three attributes

```
(root@hainguyen)-[/home/hainguyen/lab]
# echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000"
Frank manager 50000
Alex clerk 45000
Eirc clerk 25000

(root@hainguyen)-[/home/hainguyen/lab]
# |
```

- Select lines with the key word “manager”
- Select attributes 1 and 3

```
(root@hainguyen)-[/home/hainguyen/lab]
# echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000" \
| awk '/manager/ { print }'
Frank manager 50000

(root@hainguyen)-[/home/hainguyen/lab]
# echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000" \
| awk '{ print $1, $3 }'
Frank 50000
Alex 45000
Eirc 25000

(root@hainguyen)-[/home/hainguyen/lab]
# |
```

- Show Row Number

echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000" | awk '{print NR, \$1}'

```
(root@hainguyen)-[/home/hainguyen/lab]
# echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000" | awk '{print NR, $1}'
1 Frank
2 Alex
3 Eirc

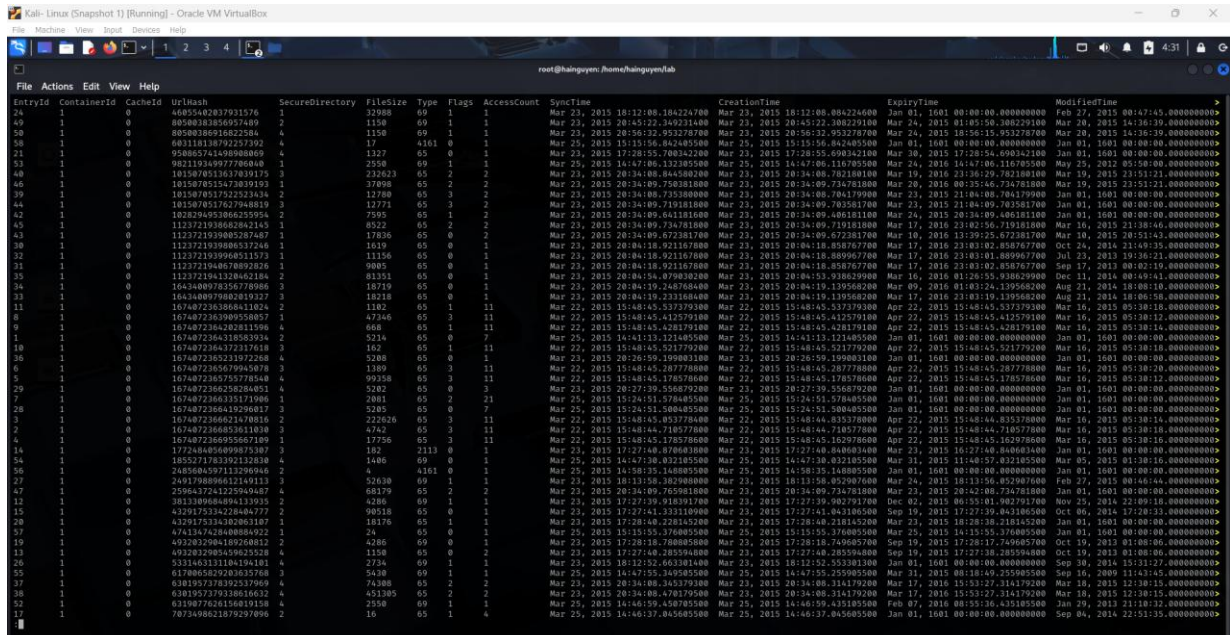
(root@hainguyen)-[/home/hainguyen/lab]
# |
```

- Calculate sum

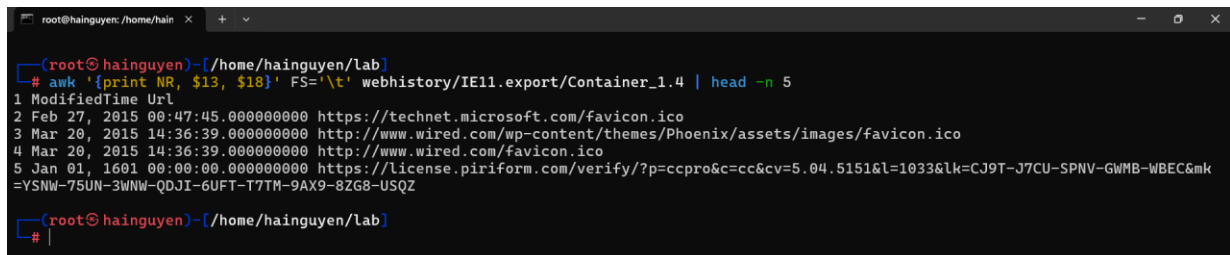
echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000" | awk '{sum += \$3} ; END {print sum}'

```
root@hainguyen: /home/hain
(root@hainguyen)-[/home/hainguyen/lab]
# echo -e "Frank manager 50000 \nAlex clerk 45000 \nEirc clerk 25000" \
| awk '{ sum += $3 } END { print sum }'
120000
```

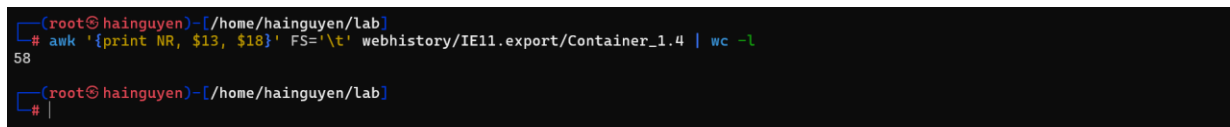
- Show field names



- ```
awk '{print NR, $13, $18}' FS='\t' webhistory/IE11.export/Container 1.4 | head -n 5
```



- ```
awk '{print NR, $13, $18}' FS='t' webhistory/IE11.export/Container 1.4 | wc -l
```



- ```
awk '{print NR, $13, $18}' FS='|' webhistory/IE11.export/Container* | wc -l
```



**Step 3.**

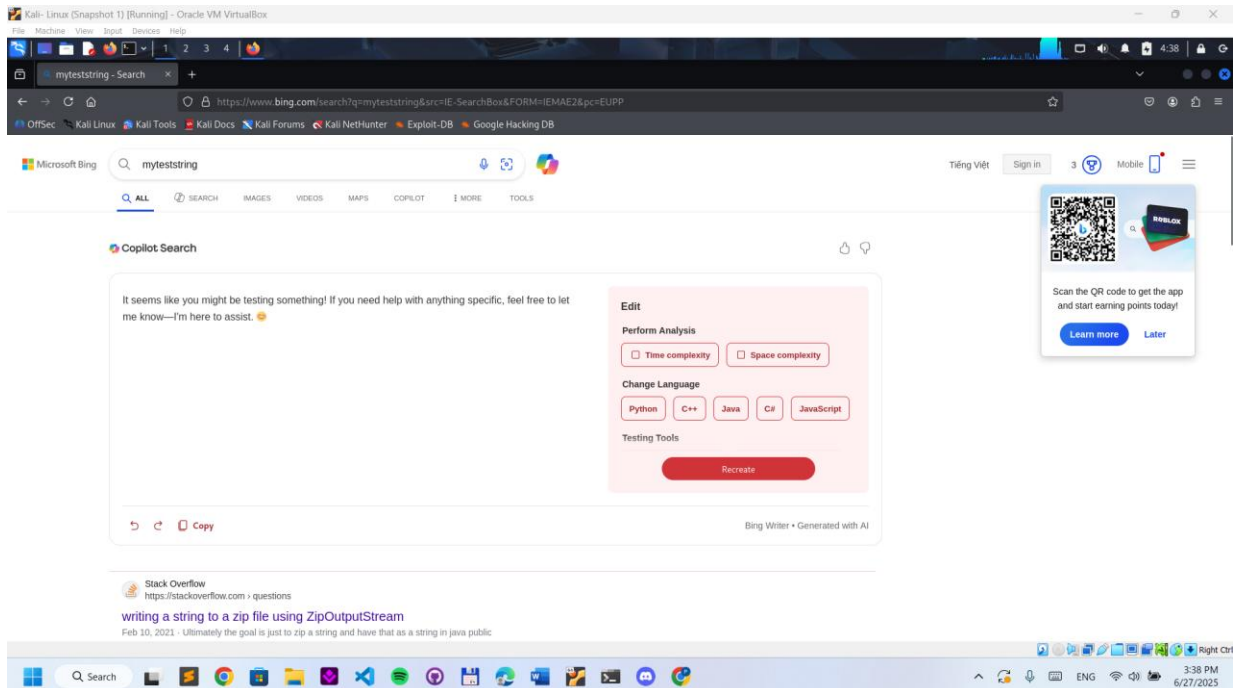


List all search keywords using web browsers. (Timestamp, URL, keyword...)

- IE browser

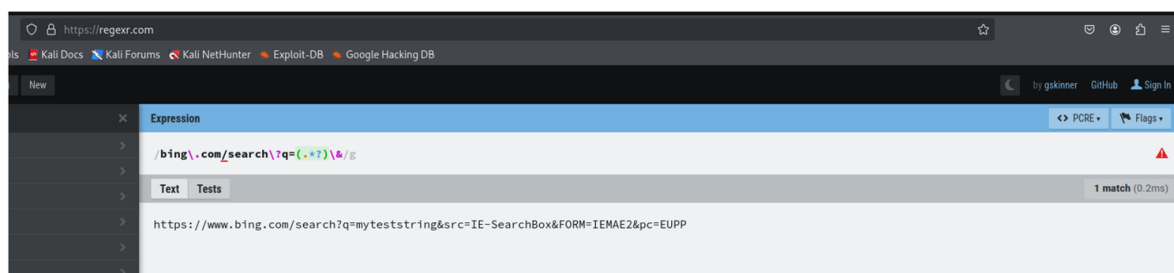
## Web Query String

<https://www.bing.com/search?q=myteststring&src=IE-SearchBox&FORM=IEMAE2&pc=EUPP>



- Regular expression for bing.com search

<https://regexr.com/>



`<http>Hello World</http>`

`<.+>`: Greedy will consume as much as possible

`<.+?>`: lazy will stop right after matching



- List IE 11 search keywords

```
(hainguyen@hainguyen) [~]
$ cd lab

(hainguyen@hainguyen) [~/lab]
$ sudo su
[sudo] password for hainguyen:
(root@hainguyen) [/home/hainguyen/lab]
$ awk '{print $34}' webhistory/IE11.export/Container* | grep -P "bing\.com/search\?q=(.*)\&" --color
informant@http://www.bing.com/search?q=Top+Stories&FORM=HDRSC1
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&sc=n&form=QBRE&ppq=external%20device%20and%20forensics&sc=8-9&sp=1&sk=&cvd=c30c4b1f36114b1c9bc683838c69823a
informant@http://www.bing.com/search?q=anti-forensic+tools&sc=n&form=QBRE&ppq=anti-forensic+tools&sc=8-13&sp=1&sk=&cvd=e799e715fa2244a5a7967675bdcca9d3
informant@http://www.bing.com/search?q=file+sharing+and+tethering&sc=n&form=QBRE&ppq=file+sharing+and+tethering&sc=0-18&sp=1&sk=&cvd=171b77e4ffd54b2a92c4e97abf995fe1
informant@http://www.bing.com/search?q=DL%20ORM&sc=n&form=QBRE&ppq=dlp%20dr&sc=8-7&sp=1&sk=&cvd=6e206ee8751e4ad89f882ed52da3aea&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsncb
bid=8
informant@http://www.bing.com/search?q=what%20is%20windows%20system%20artifacts&sc=n&form=QBRE&ppq=what%20is%20windows%20system%20artifacts&sc=0-27&sp=1&sk=&cvd=1ef4ace146854d97acf263b53b
f97b8&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsncbid=4
informant@http://www.bing.com/search?q=Top+Stories&FORM=HDRSC1
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&sc=n&form=QBRE&ppq=external%20device%20and%20forensics&sc=8-9&sp=1&sk=&cvd=c30c4b1f36114b1c9bc683838c69823a
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&sc=n&form=QBRE&ppq=external%20device%20and%20forensics&sc=8-9&sp=1&sk=&cvd=c30c4b1f36114b1c9bc683838c69823a&sid=
BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsncbid=9
informant@http://www.bing.com/search?q=Forensic+Email+Investigation&FORM=QSR&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsncbid=3
informant@http://www.bing.com/search?q=cd%20burning%20method&sc=n&form=QBRE&ppq=cd%20burning%20method&sc=8-2&sp=1&sk=&cvd=b7db6fb67424c578172ba57330a0894&sid=BE5E388F8757406CAA32E5833471
9A20&format=jsonv2&jsncbid=7
informant@http://www.bing.com/search?q=mail+investigation&sc=n&ppq=mail+investigation&sc=8-7&sp=1&sk=&cvd=felc3738d8c7471284731724166959af&first=9&FORM=PERE&sid=BE5E388F8757406CAA32E5
8334719A20&format=jsonv2&jsncbid=2
informant@http://www.bing.com/search?q=investigation%20on%20windows%20machine&sc=n&form=QBRE&ppq=investigation%20on%20windows%20machine&sc=8-4&sp=1&sk=&cvd=eb73de7f523c48769d56281379f55e6
7&sid=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsncbid=5
informant@http://www.bing.com/search?q=eraser&sc=n&form=QBRE&ppq=eraser&sc=8-6&sp=1&sk=&cvd=e3b983f889944179893ff5199b2eac4&sid=C7EBF3776E804120B57C623F21EF33C4&format=jsonv2&jsncbid=0
informant@http://www.bing.com/search?q=windows%20event%20logs&sc=n&form=QBRE&ppq=windows%20event%20logs&sc=0-32&sp=1&sk=&cvd=36b33ac5151246398f7d1ca79de069c&sid=BE5E388F8757406CAA32E5833
4719A20&format=jsonv2&jsncbid=6
informant@http://www.bing.com/search?q=anti-forensic+tools&sc=n&form=QBRE&ppq=anti-forensic+tools&sc=8-13&sp=1&sk=&cvd=e799e715fa2244a5a7967675bdcca9d3
informant@http://www.bing.com/search?q=cd%20burning%20method&sc=n&form=QBRE&ppq=cd%20burning%20method&sc=0-8&sp=1&sk=&cvd=acec9b1dcb8146c58258ad65c770d76e&si
d=BE5E388F8757406CAA32E58334719A20&format=jsonv2&jsncbid=8
informant@http://www.bing.com/search?q=cclleaner&sc=n&form=QBRE&ppq=cclleaner&sc=8-8&sp=1&sk=&cvd=dl34736d4e514ad497f68734a6779104&sid=C7EBF3776E804120B57C623F21EF33C4&format=jsonv2&jsncbi
d=1
informant@http://www.bing.com/search?q=file+sharing+and+tethering&sc=n&form=QBRE&ppq=file+sharing+and+tethering&sc=0-18&sp=1&sk=&cvd=171b77e4ffd54b2a92c4e97abf995fe1
informant@http://www.bing.com/search?q=Top+Stories&FORM=HDRSC1
informant@http://www.bing.com/search?q=external%20device%20and%20forensics&sc=n&form=QBRE&ppq=external%20device%20and%20forensics&sc=8-9&sp=1&sk=&cvd=c30c4b1f36114b1c9bc683838c69823a
informant@http://www.bing.com/search?q=file+sharing+and+tethering&sc=n&form=QBRE&ppq=file+sharing+and+tethering&sc=0-18&sp=1&sk=&cvd=171b77e4ffd54b2a92c4e97abf995fe1
informant@http://www.bing.com/search?q=anti-forensic+tools&sc=n&form=QBRE&ppq=anti-forensic+tools&sc=8-13&sp=1&sk=&cvd=e799e715fa2244a5a7967675bdcca9d3

(root@hainguyen) [/home/hainguyen/lab]
$
```

**YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!**

Save the document with the filename "**YOUR NAME Lab 11.pdf**", replacing "**YOUR NAME**" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 11 From YOUR NAME**", replacing "**YOUR NAME**" with your real name.